

Week 11 Lesson Plan

The Architecture of AI Agents — Four Arenas, One Law

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Four Arenas, One Law

Any organization that designs a system (defined broadly) will produce a design whose structure is a copy of the organization's communication structure.

Melvin Conway, *How Do Committees Invent?* (1968)

Be conservative in what you do,
be liberal in what you accept
from others.

Jon Postel, the Robustness Principle, RFC 761 (1980)

A Gentle Re-Entry

FOR TWO WEEKS THE HARNESS HELD THE STAGE. We learned that the model computes but the machine around it ships; that a system without a feedback loop is a gesture; that the loop's four organs — assemble, propose, act, verify-and-feed-back — are where reliability is manufactured, not found. Today we ask the question the harness weeks deliberately postponed: *how many loops, and how arranged?* One agent or many? A star, a pipeline, a committee, a swarm? This is the week of **architecture** — and it opens not with a taxonomy but with a provocation.

HERE IS THE PROVOCATION. The arithmetic of the open loop is merciless. A step that succeeds with probability p , run n times without correction, succeeds with probability p^n ; at ten steps and ninety percent, that is $0.9^{10} \approx 0.35$, and at a *million* steps even four nines collapse to nothing:

Hold the number in your head all morning: a million-plus consecutive steps, each proposed by a fallible model, and not one surviving error. Either the arithmetic is wrong, or the arithmetic is describing a different machine than the one they built.

$0.9999^{1,000,000} \approx 0$. By that arithmetic, long-horizon agentic work is impossible. And yet: a research team has run a task of over a million LLM steps — the twenty-disk Tower of Hanoi, $2^{20} - 1$ moves — with *zero* errors. How in the world is that possible? And why is everyone and their best friend drooling over agentic systems for every emerging task, when the naive math says the drool should dry up by step thirty? We will not answer this now. You will answer it yourselves — four times, in four arenas, before lunch. By the close of the day you should be able to say precisely *when* the enthusiasm is rational, and when it is admiration for an open loop that has not failed *yet*.

AND THERE IS A CRUELER VERSION of the provocation, which we must plant early because it shapes everything you design today. The p^n curve at least *decays* — you can buy more nines, add retries, and claw back probability. But suppose an early step concludes something *wrong*, and every subsequent step treats that conclusion as gospel truth: never re-examined, only built upon. From that moment the probability of success is not small. It is exactly **zero**, no matter how brilliant the remaining steps are. We will call this the **fatal move**. The million-step run is astonishing twice over: it had a million opportunities to make one fatal move, and it made none. Ask of every design you draw today: *when step one is wrong, who can find out — and how does the system un-believe it?*

What Must You Carry Forward

WHEN TODAY'S SESSION CONCLUDES, you should be able to:

- State the law of the day — *architecture follows the information* — and apply it: given a task, say where the information lives and let that dictate the topology.
- Name the two, and only two, honest justifications for multi-agent designs: private knowledge distributed among parties, and context capacity exceeded by the world.
- Compute open-loop decay p^n for a given chain, and explain why a verifier with retries converts a 35% one-shot generator into an 80%-plus system.
- Define the fatal move, distinguish it from ordinary decay, and name the structural cure: a loop that can reach *back* and un-believe an earlier conclusion.

- Draw the four harness organs — Assembly, Proposal, Action, and Verification with its feedback — onto any topology, and identify a drawing with no verifier as p^n in costume.
- Recite the workhorse patterns — pipeline, router, star orchestrator, generator-verifier, fan-out/reduce — with the condition under which each is optimal.
- Explain the dominance criterion from the recent single-versus-multi-agent science: extra agents pay only when they contribute *exogenous* information the first agent lacked.
- Describe error amplification in badly coupled multi-agent systems, and why a committee can be confidently, politely, and unanimously wrong.
- Give the characteristic fatal move of each of today's four arenas, and where the winning design placed its verifier.
- State von Neumann's 1956 result on building reliable machines from unreliable components, and say in one sentence why it licenses the whole agentic enterprise.
- Explain TCP over IP as the reliability theorem you have lived: an unreliable substrate plus a closed loop equals a reliable system, with nothing in the wire improved.
- Say why a role-play leak log is a formal audit of a topology: a human breaks role exactly where the design starved their agent.
- Argue that a well-written skill is an architectural component — a contract that makes delegation trustworthy — using the Toast verifier and the Wikipedia researcher as cases.
- Run the architect's catechism: the ordered questions that take you from a task statement to a defensible topology, without ever reaching for fashion.

If you can do these without hand-waving, the afternoon's catalog will feel like naming old friends rather than memorizing strangers.

The Shape of the Day

THE MORNING BELONGS TO YOUR HANDS; the afternoon belongs to names. You will be six groups. Each group meets *two* of four problems — once as **human role-play**, once as an **ADK build** — and each problem is met by exactly three groups, so that every arena ends the morning with a comparison panel: same brief, three topologies, side by side. That comparison, not any single build, is the raw material of the debrief.

Group	Round 1 · role-play	Round 2 · ADK build
G1	P1 The Potluck	P2 The Symposium
G2	P2 The Symposium	P3 The Toast
G3	P3 The Toast	P4 Vacation, One Rung Up
G4	P4 Vacation, One Rung Up	P1 The Potluck
G5	P1 The Potluck	P3 The Toast
G6	P2 The Symposium	P4 Vacation, One Rung Up

Block	Duration	What happens
Opening brief	10 min	the provocation; the rules; the four organs; fatal moves
Round 1 · role-play	25 min	5 min to draw the topology, 20 to enact it
Round 2 · ADK build	55 min	build the second problem; freeze at the 10-min warning
Debrief	35 min	arena by arena: three groups compared, verdicts unsealed
<i>Lunch</i>	—	<i>the verdicts settle</i>
Afternoon	—	the catalog named; the science; the architect's procedure

THE RULES OF THE GAME ARE THREE. First, the briefs are deliberately vague about *how* — they specify deliverables and grading, never topology; the diversity of your designs is the curriculum. Second, before you build or enact anything, you *draw* your topology — boxes with one-line roles, arrows for who may talk to whom — and you label where the four harness organs live on the drawing. The organs may be fused, split, or delegated; but if the verifier cannot be found, you have drawn p^n , and the graders are merciless. Third, once enactment starts, the drawing is *law*.

The Opening Brief, Verbatim

THE MORNING'S FRAMING DESERVES TO BE DELIVERED IN NEARLY FIXED WORDS, because every phrase is load-bearing. After the provocation — the impossible number, and its crueler cousin — the room hears approximately this:

“Six groups, two problems each. Round one, you don’t touch a computer: you design an agent topology on paper, then *become* it — one of you is the harness, the runtime, and enforces the drawing ruthlessly. Round two, you build your other problem in ADK. The briefs are deliberately vague about *how*. There are hidden grading keys. At the end, three groups will have attacked each problem — we’ll put your designs side by side and see which architectures earned their keep. One rule: your topology drawing is law. If you feel the urge to break it mid-run — notice that urge. It is the most important data of the morning.”

“And one requirement, from the harness weeks: p^n is a disaster, and it is what open loop looks like. So every topology you draw must *label* where the four harness elements live: context Assembly, LLM Proposal, Tool Action, and Verification-of-result feeding back. They need not be four boxes — they may be fused, split, or delegated — but if I can’t find your verifier in the drawing, you have drawn p^n , and the graders are merciless. Last thing, the deadliest: beware the fatal move. The moment an early step concludes something wrong and the rest of your system treats it as gospel — never questioned, only built upon — your probability of success is zero. Not small: zero. So ask of your design: when step one is wrong, *who* can find out, and *how* does the system un-believe it?”

Nothing in either paragraph names a topology, and nothing may: the requirement tells every group to close the loop, but not *where* the verifier belongs for their problem — that placement differs per arena, and discovering it is the morning’s work.

Becoming the Machine: the Role-Play Protocol

IN THE ROLE-PLAY ROUND, ONE OF YOU IS NOT AN AGENT. You are the **harness** — dumb infrastructure, and proudly so. Agents may not speak. All communication is written notes, handed to the harness, delivered only along a drawn arrow; a note with no arrow is returned undelivered. A note that runs past about twenty-five words is torn at the fold and only the first half delivered — that is what context truncation feels like from the inside. The harness may not summarize, hint, repair, or route creatively.

THE HARNESS KEEPS ONE ARTIFACT that will outlive the morning: the **leak log**. Every time a human breaks role — speaks aloud, uses knowledge from another card, applies common sense their one-line instruction never granted, quietly changes their own role mid-run — the harness writes a line: who, what leaked, and the best guess at *why the topology made them do it*. Here is the reframe that makes the log precious rather than punitive: *leaks are data*. A human breaks role at

When a Round-1 leak in one room — “no-body ever asked about money” — matches a Round-2 silent failure in another room — the built Potluck violating the hidden budget constraint — read the pair aloud. There is no better single moment in the debrief.

exactly the point where the agent they are playing was starved — of information, of authority, or of a channel. LLM agents fail in the same places; they simply fail *silently* instead of cheating. Your leak log is a free failure-mode audit of your own architecture, run with humans as the model checker.

The Four Organs, Formally

BEFORE THE ARENAS OPEN, WE RESTATE THE HARNESS INHERITANCE in the notation that today’s drawings must honor. One revolution of any loop, however many agents share it, is:

$$c_t = A(s_t) \quad a_t = \pi_\theta(c_t) \quad o_t = W(a_t) \quad s_{t+1} = V(s_t, a_t, o_t)$$

Assembly A chooses what this step gets to see; the model π_θ proposes; the world W answers; and verification V decides what the observation *means* before folding it into state. The model owns exactly one box. Architecture, in this notation, is the set of decisions about how many copies of this loop exist, which boxes they share, and whose V is allowed to overwrite whose s .

Notice what the equations quietly assert: nothing enters the state s_{t+1} except through V . A topology that lets a proposal write directly to state — $s_{t+1} = f(s_t, a_t)$ with no o_t and no V — has built the fatal-move machine in one line.

THE FATAL MOVE ALSO HAS A ONE-LINE FORMALIZATION, and it is worth writing on the board. Call a belief *load-bearing* if later steps condition on it. The fatal move is a wrong belief entering s *without a return path*: no later V ever revisits it, so its posterior probability of correction is zero regardless of how much evidence the world subsequently offers. The cure is structural, not motivational: verification must be able to reach *back* — ratification rounds, evidence demanded per claim, records where nothing unverified is load-bearing. When you label the four organs on your drawing this morning, the grader’s second question — after “where is V ?” — is “which arrows let V un-write s ?”

Pop Quiz — the Confident Intern

TRY THIS BEFORE THE ARENAS OPEN; we will collect answers by show of hands. A ten-step pipeline runs with each step 90% reliable. In run A, the steps are merely fallible — errors happen and propagate as noise. In run B, step two states a wrong “fact” which steps three through ten faithfully build upon. Compute the success probability of run A. Then say back, in one sentence, why run B’s number is not merely *smaller*.

Answer. Run A: $0.9^{10} \approx 0.35$ — decay, ugly but survivable with retries and a verifier at the end. Run B: **zero** — the fatal move already destroyed the answer, and perfection downstream is loyalty to a false world. Decay you fix by adding nines; a fatal move you fix only with a loop that can reach back and un-believe. The two failures live in different categories, and conflating them is how open-loop demos get funded.

Act I: The Four Arenas

EACH ARENA IS A PROBLEM SMALL ENOUGH FOR A MORNING and sharp enough to make one architectural condition impossible to ignore. Do not announce the conditions — the briefs are vague on purpose, and one of the four is a trap. What follows records, for each arena, the brief as the groups receive it, what we expect to see, and what the debrief must draw out.

P1 — The Potluck: information distributed among parties

THE BRIEF. Six old friends hold a reunion potluck. Deliver a complete dinner plan: a menu (who brings what), a seating arrangement around one round table, and a start time. Each guest has told *their own agent* some private things they have not told the others — sealed role cards carry an allergy, a new celiac diagnosis a guest is embarrassed about, a feud that forbids adjacent seating, a tight budget behind a proud deflection, a babysitter deadline, a lactose intolerance joined to a hatred of cilantro. The plan is graded against a hidden eight-point checklist; every violated secret costs a point. The personas *deflect* generic questions: ask David vaguely what he will bring and he promises his famous sourdough — which he can no longer eat.

WHAT WE EXPECT. Last cohort's vacation exercise predicts the spectrum: a free-for-all multi-agent chat, charming and leaky; a star orchestrator that asks one generic "any preferences?" round and misses everything the personas deflect; and — the winning shape — an orchestrator running *specific* elicitation (allergies? budget? timing? seating?) with follow-up probes, then a **ratification round** in which the draft plan is re-presented to every guest for veto before it ships. A single agent fed only the public brief will score two to four of eight: a sensible generic plan happens to pass some checks, and that partial credit is itself a lesson about how plausible outputs conceal absent information.

WHAT THE DEBRIEF DRAWS OUT. The information was *distributed among parties*. No cleverness recovers a fact the system was never told, so elicitation is not conversational style — it is architecture. The four organs land unusually here: the tool action *is* asking the guests; the verifier *is* the guests, ratifying. The characteristic fatal move: drafting the menu *before* elicitation finishes and thereafter only tweaking it — from that moment David’s celiac disease can never enter the system, and the plan’s fate was sealed while it still looked half-done.

P2 — The Symposium: the trap

THE BRIEF. Five speakers, five slots around a lunch break, eight hard constraints — all printed in full, nothing hidden, exactly one valid schedule. Deliver the schedule, and say why you *trust* it rather than merely like it.

WHAT WE EXPECT — AND WHY THIS ONE IS THE TRAP. At least one group will build a negotiation: five speaker-agents bargaining for slots, or a committee voting in rounds. It will be slow, and it will often be *wrong with confidence* — agents concede constraints politely, a tentative early placement (“let’s fix Rao in slot one and go from there”) is quietly promoted to gospel, and consensus inertia carries the error home. Groups that use one agent with a check-your-work loop will get it right. A group that writes twenty lines of code to search all 120 permutations wins outright, and should be praised loudly: that is *code-as-orchestration*, the most honest architecture on offer, and the model’s proper job here is at most to *write* the search, not to *be* the search. In the role-play rooms this arena is comedy: watch for the human who privately solves the whole puzzle in ninety seconds and must then smuggle the answer through a negotiation topology — capture it in the leak log; it is the dominance criterion wearing a party hat.

WHAT THE DEBRIEF DRAWS OUT. Ask the committee-builders one question: *what information did any second agent bring that the first agent lacked?* The silence is the lesson. Everything was in the brief; nothing was distributed; every extra agent was pure error surface. Then deliver the whiplash, which is the deepest single sentence of the morning: *the architecture that won the Potluck is the architecture that lost the Symposium*. Same morning, same people, same patterns — the problems differ in exactly one property, which is where the information lives.

P₃ — The Toast: success that is checkable

THE BRIEF. Write the best man’s toast for Anika and Rohan — grounded in a supplied fact sheet (the climbing-gym rescue, the food-truck down-pour, a terrier named Biscuit with theatrical dining requirements) — and pass a *published* eight-item rubric: at most 150 words; both names; exactly one joke; none of the banned clichés; at least one specific shared memory; Biscuit exactly once; a final sentence beginning “To”; at most one exclamation mark. Six of the eight checks are graded mechanically by a supplied script; the joke and the memory require a judge, human or LLM. The deliverable includes the iteration count.

WHAT WE EXPECT. One-shot generation plus manual patching becomes whack-a-mole: fixing the word count breaks the Biscuit count; trimming the cliché kills the joke. A committee of style agents debates taste while the word count sits at 212. The winning shape is the **generator-verifier loop**: generate, run the mechanical gate, feed back *the failed check identifiers* — not vibes — regenerate, and pay for the judge only on candidates that clear the free checks. Convergence typically lands in two to four iterations. Watch for the reward hacker: a toast that satisfies all eight constraints and has no warmth in it. Name it kindly — *you optimized the rubric, not the toast* — and let Goodhart take a bow.

WHAT THE DEBRIEF DRAWS OUT. The verifier arithmetic, concretely: a generator that passes all eight one-shot maybe a third of the time becomes, behind a reliable checker with a handful of retries, a system that ships north of eighty percent — checking is cheaper than being right the first time. And the rubric’s own split teaches the production cascade: cheap deterministic gates *first*, expensive judgment *last*. The characteristic fatal move: locking the draft’s structure after iteration one and thereafter only patching words, when the structure itself cannot satisfy the joke and the dog at once — feedback must be allowed to *kill* a draft, not merely amend it.

P₄ — Vacation, One Rung Up: information out in the world

THE BRIEF. Last cohort planned a vacation by negotiation — the information lived in the family. This time the world pushes back. Twelve candidate destinations; a six-item family rubric (total flying time from San Francisco, late-December weather, activities for children of eight and eleven, real hiking, easy vegetarian eating, mid-range budget); and

one iron rule: every factual claim must be grounded in that destination's Wikipedia page, reached through the supplied `wikipedia_lookup` tool. Deliver a ranked top three with per-item verdicts and the page evidence for each verdict. We will spot-check; a claim not on the page is a hallucination and disqualifies the destination.

WHAT WE EXPECT. The design fork we are waiting for has two prongs. Who reads: one agent trudging through twelve pages sequentially — context bloats, later verdicts get sloppy, and facts from Kyoto begin bleeding into Lisbon — or a fan-out of parallel researchers, one page each? And who judges: researchers grading *their own* destination — self-grading drifts lenient; “Bali is probably fine in December” when the page plainly says wet season — or researchers producing grounded digests only, while an independent judge holds the rubric, demands a quotation per verdict, and owns a veto that sends a researcher back? The menu is salted with traps that reward actual reading: the wet season, a flight budget that quietly disqualifies the southern-summer paradise, four hours of December daylight in one candidate, altitude with children in another.

WHAT THE DEBRIEF DRAWS OUT. The winning shape is fan-out plus an evidence-demanding judge — *separation of duties* as an architectural principle. The characteristic fatal move is the most production-realistic of the four: a hallucinated “fact” enters a digest, and the judge treats digests as ground truth; everything downstream is faithful reasoning about a false world, and only quotation-level verification can catch it. And the closing question ties the morning's law together: last cohort's vacation needed many agents because the information lived in the *family*; today it lived in *Wikipedia*. Same fan-out shape — name the two different reasons. Private knowledge, and context capacity. Those two, and nothing else, are the honest justifications for a society of agents.

A Gallery of Expected Topologies

THREE GROUPS PER ARENA MEANS THREE DRAWINGS PER ARENA, and experience with the vacation exercise lets us sketch the gallery in advance. Treat these as predictions to check, not prescriptions to leak — the debrief gains its force precisely when the room discovers it reproduced the gallery unprompted.

Potluck, naive. Six guest agents in a free-for-all group chat, three or four rounds, no chair. Charming transcript, warm feelings, and a

plan that misses whatever nobody happened to blurt out — typically the budget and the start time, because neither is dinner-table conversation. Organs: assembly everywhere, verification nowhere.

Potluck, better. A star: orchestrator asks each guest once, integrates, ships. The generic single pass falls to the deflecting personas — David still brings bread. The drawing has a verifier box, but its arrow points only forward.

Potluck, winning. The star with two amendments: *specific* elicitation with follow-ups, and a ratification arrow looping the draft plan back through every guest before it ships. Verification by the constraint-holders themselves; the fatal move structurally excluded.

Symposium, naive. Five speaker agents negotiating slots — the org-chart reflex in its purest form. Slow, polite, frequently wrong, and wrong together.

Symposium, better. One agent, chain-of-thought, then a self-check pass over the eight constraints. Usually right; trust rests on the check, not the reasoning.

Symposium, winning. Twenty lines of code enumerating 120 permutations, the model at most writing the code. Proof, not confidence. The debrief's question to the room: which organ was the `for` loop? (Proposal and Verification, fused — and that fusion is why it cannot make a fatal move.)

Toast, naive. One-shot generation, human eyeball, manual patching — whack-a-mole with a word budget.

Toast, better. A taste committee debating warmth while the mechanical checks fail unattended; verification aimed at the wrong altitude.

Toast, winning. Generator, mechanical gate, failed check identifiers fed back, judge reserved for survivors — the cascade with the cheap checks in front.

Vacation, naive. One agent reads twelve pages in sequence; by page nine the digests blur and Kyoto's temples acquire Lisbon's trams. Context, not diligence, is the binding constraint.

Vacation, better. Fan-out with self-grading researchers — clean rooms, but each grading its own homework, drifting lenient exactly where the rubric pinches.

Vacation, winning. Fan-out to digest-only researchers under the skill contract, one skeptical judge holding the rubric and the veto, quotations demanded per verdict. Separation of duties drawn as arrows.

Materials, Mechanics, and the Facilitator's Eye

THE KIT LIVES IN `agents-course/archcraft-morning/`: printable briefs and role cards under `handouts/`, grading keys and the unique Symposium solution under `ta-only/` (facilitators only — the trap survives exactly one leak), and code under `kit/`: a mechanical grader for the Symposium, the toast checker, and the `wikipedia_lookup` tool with its on-disk cache. Three mechanics matter enough to rehearse. First, *pre-warm the Wikipedia cache* on the classroom network before the sprint — one command, thirty seconds — so that Arena Four survives any mid-morning network sulk and all groups research against identical snapshots, which keeps the debrief comparisons honest. Second, the ADK crib sheet in the handouts covers framework ceremony only — imports, the agent constructors, the composition primitives — and deliberately contains no topology hints; a group stuck ten minutes on plumbing gets rescued immediately, because plumbing is not the lesson, while a group asking “should we use an orchestrator?” gets exactly one sentence back: *what does the problem's information structure say?* Third, scope-cutting is a kindness: at the sprint's two-thirds mark, any group with nothing running is told to shrink — fewer guests, five destinations instead of twelve, mechanical checks only — because a small thing that runs teaches more than a grand thing that does not.

FOUR SAY-IT-BACK CHECKPOINTS, one per arena, to be asked while circulating — not to steer, but to make groups articulate their own design out loud:

- *Potluck*: which question in your protocol would surface a secret its holder is embarrassed about — and what happens to your plan if no question does?
- *Symposium*: finish this sentence honestly — “we will trust our schedule because...” If the sentence ends in “the agents agreed,” start over.
- *Toast*: when a draft fails three checks, what *exactly* travels back to the generator — the verdicts, the vibes, or the check identifiers?

- *Vacation*: if one researcher quietly invented an amenity, name the component in your design that would catch it — and what it would cost to run.

Act II: The Catalog, Named

THE AFTERNOON BEGINS BY NAMING WHAT THE MORNING BUILT. Every topology you drew is a point in a small, well-charted design space. We climb it as a ladder, and at every rung we ask the same two questions: *where does the information live?* and *where does the loop close?*

The Ladder

Rung 0 — one call. A single model call, no tools, no loop. A gesture. Fine for what fits in one head and needs no checking; the p^n arithmetic never starts because $n = 1$.

Rung 1 — one agent, closed loop. The harness weeks' machine: assemble, propose, act, verify, feed back. This is the *default* architecture, and the baseline every fancier design must beat. The Symposium belongs here — or one rung down, in plain code.

Rung 2 — one agent, many instruments. Still one mind, but the verifier diversifies: mechanical gates, test suites, a judge model at the end. The Toast lives here: one generator, a cascade of checks, a loop that feeds failures back by name.

Rung 3 — many agents, one law. A society — star, pipeline, fan-out, peer team — justified *only* by information a single agent cannot reach: sealed cards (the Potluck), or twelve pages no single faithful reading can hold (the Vacation).

The Workhorse Patterns

FIVE PATTERNS DO ALMOST ALL THE HONEST WORK in deployed systems. Each is optimal under a condition you can state in one line, and each failed visibly this morning wherever its condition did not hold.

The pipeline. Stages in fixed order, each consuming the last stage's output. Optimal when the task decomposes cleanly and stage boundaries are natural checkpoints. Its besetting sin is the fatal

move: stage N treating stage $N-1$'s output as gospel. A pipeline without inter-stage verification is p^n drawn horizontally.

The router. One triage step dispatching to specialists. Optimal when inputs are heterogeneous and specialists genuinely differ. The router's failure is quiet misclassification — a fatal move at step zero, dooming everything downstream while every specialist performs beautifully.

The star orchestrator. A hub that elicits, delegates, integrates — the Potluck's winner. Optimal when information is distributed among parties and someone must own the global picture. Its fatal move: the hub that stops eliciting and starts assuming.

The generator-verifier loop. The Toast's winner, and the verifier arithmetic's home. Optimal when success is checkable more cheaply than it is achievable. The order of checks is part of the architecture: deterministic gates first, judgment last.

Fan-out / reduce. Parallel workers over a partition, one reducer with the global view — the Vacation's winner. Optimal when the world exceeds one context and partitions cleanly. Its integrity rests entirely on the reducer's skepticism: a credulous reducer inherits every worker's hallucination at once.

The Adaptive Fringe — and a Reframe

BEYOND THE WORKHORSES LIES THE ADAPTIVE FRINGE: debate and peer teams, blackboard architectures where agents rendezvous on shared state, hierarchical planners spawning workers, and *code-as-orchestration* — the Symposium's quiet champion — in which the control flow is written in a programming language and the model is invited in only where judgment is genuinely needed. Treat the fringe with respect and suspicion in equal measure: every pattern here has a demo in which it shines and a benchmark in which it loses to Rung 1 plus a good verifier.

A committee is not a truth machine. Debate helps when arguers hold *different evidence*; it merely launders confidence when they hold the same context — polite agreement is not verification, a lesson the Symposium's negotiators will have felt in their bones.

AND ONE REFRAME TO CARRY: much of what is sold as “multi-agent” is better understood as **context partitioning** — not a society of minds but a filing discipline, keeping each sub-task's working set clean so that attention is spent where it belongs. The Vacation's researchers are not colleagues; they are separate clean rooms. Seeing partitioning as the point — rather than personhood — dissolves most of the romance and most of the confusion at once.

The Protocol Layer, in One Breath

ABOVE THE TOPOLOGY SITS A FOURTH LAYER, and the harness weeks earned us the right to name it briefly: the **protocol layer**. Inside one system, your architecture is whatever you draw. Between systems — an agent calling tools it did not ship with, an agent conversing with an agent it has never met — the arrows must be *standardized*, and the field has spent the past two years standardizing them: a common protocol for mounting tools and context onto a model (the tool socket), and a common protocol for agent-to-agent exchange (the envelope between societies). The details belong to the MCP week and to Archcraft's protocol chapter; today only the architectural moral matters. Protocols are the TCP of the agent world: they assume the other side is imperfect and wrap correction around the exchange — typed contracts, explicit capabilities, refusals as first-class replies. Postel again: conservative in what you emit, liberal in what you accept. An architecture whose components assume each other's perfection is a fatal-move machine with good manners; a protocol is institutionalized suspicion, and it is the reason strangers' components can be composed at all.

Act III: The Science and the Law

THE MORNING GAVE YOU INSTANCES; THE SCIENCE GIVES YOU THE LAW. Three results — one piece of arithmetic, one theorem-shaped criterion, one anatomy of failure — plus two old friends from outside the field who settle the provocation.

The Reliability Calculus

OPEN LOOP: success p per step, n steps, no correction — p^n , the number that opened the day. Worse, real models *self-condition*: a transcript containing errors makes further errors more likely, so the effective p *falls* as the run lengthens; the open-loop curve is optimistic.

CLOSED LOOP: let a verifier catch failures and allow retries. A generator that is right 35% of the time, gated by a reliable checker with a few retries, ships north of 80%: with k independent attempts, the chance that *none* passes the gate is $(1 - 0.35)^k$ — at $k = 4$, about 0.18 — and the verifier ensures that what ships is drawn from the attempts that

The verifier's own error matters, and asymmetrically: a *false-positive* verifier — one that blesses wrong work — is worse than none, because it manufactures gospel. The harness week's lying round of warmer/colder was the felt version; the Vacation's credulous judge is today's.

passed. The exponent that killed the open loop now works for you: retries compound success instead of failure.

AND THE MILLION STEPS? The team behind the Tower of Hanoi run — the MAKER result — did two things with total discipline: they *decomposed* the task until each step was tiny and trivially checkable, and they *verified every step*, red-flagging suspect outputs and letting independent samples vote, first-to- k . Two consequences follow. The effective per-step reliability is pushed so close to one that even a million-step exponent cannot bite. And — the part that answers the crueler provocation — no wrong step ever *survives into the record* to become gospel: the fatal move is not merely discouraged but structurally impossible, because nothing unverified is ever built upon.

Horizons, and Why the Drool Is Half Right

THE SAME CALCULUS EXPLAINS THE FIELD'S GENUINE EXCITEMENT, and it is worth being fair to it. The task horizon an agent can complete reliably has been *doubling* on a timescale of months — not because raw per-step brilliance doubles, but because harnesses, verifiers, and decomposition discipline keep converting fallible steps into correctable ones. When the loop is closed, the scaling *flips*: instead of p^n eating the task, every improvement in per-step reliability or verification coverage compounds across the whole horizon. A concrete feel for the stakes:

Steps n	$p = 0.90$	$p = 0.99$	$p = 0.999$
10	0.35	0.90	0.99
100	2.7×10^{-5}	0.37	0.90
1,000	≈ 0	4.3×10^{-5}	0.37
1,000,000	≈ 0	≈ 0	≈ 0

Read the table twice. Downward: no fixed p survives a long horizon — more nines only delay the cliff, so *open-loop scaling is hopeless at every quality level*. Across: at any fixed horizon, nines help enormously — which is what per-step verification buys. The million-step run lives *off* this table entirely, because per-step correction makes the effective exponent stop mattering. That is the whole story of the drool: rational wherever the loop closes, delusional wherever someone is quietly reading a row of this table and calling it a roadmap.

Self-conditioning makes the open-loop story worse than the table: a transcript containing errors makes further errors *more* likely, so p itself decays with n . The open loop is not a constant-hazard machine; it is a machine whose hazard grows as it tires.

Von Neumann's Theorem, and the Theorem You Have Lived

NONE OF THIS IS NEW UNDER THE SUN. In 1956, John von Neumann published *Probabilistic Logics and the Synthesis of Reliable Organisms from Unreliable Components*: with redundancy and majority voting, a machine built from components that each err with probability ε can be made to compute with error as small as you please, provided ε sits below a threshold. Reliability, the theorem says, was never a property of the component; it is a property of the *correction architecture* wrapped around the component. Noisy transistors; exact laptops. Fallible LLM steps; an exact million-step run.

AND YOU HAVE TRUSTED THIS THEOREM YOUR ENTIRE CAREER, every time you opened a socket. IP is unreliable: packets drop, packets corrupt, packets arrive out of order. TCP, built *on top of* IP, is reliable: checksums, acknowledgements, retransmission, sequence numbers. Nothing about the wire ever improved. The magic is in the loop — and note the direction of the construction: reliability is built *on top of* a fallible layer, never *into* it. That is exactly what a harness is to a model, and what today's architectures are to a harness.

Postel's epigraph belongs to this layer: protocols survive because each side assumes the other is imperfect and corrects for it. An architecture whose components assume each other's perfection is a fatal-move machine with good manners.

The Whiplash, Formalized

PUT THE POTLUCK AND THE SYMPOSIUM SIDE BY SIDE and the day's law falls out as a controlled experiment — two problems identical in size, effort, and glamour, differing in one bit:

	P1 Potluck	P2 Symposium
Constraints	eight, <i>hidden</i> , held by parties	eight, <i>printed</i> , held by no one
Reachable by one agent?	no — must be elicited	yes — all in the brief
Extra agent brings...	a secret nobody else holds	nothing but error surface
Winning topology	star + elicitation + ratification	one agent + check, or plain code
Committee's fate	necessary — <i>it is</i> the source	confidently, politely wrong

Same morning, same students, same patterns on the whiteboard — and the optimal architecture inverts on a single property. Any student who leaves today able to reconstruct this table from memory has the entire week; everything else is elaboration.

CONWAY'S EPIGRAPH NOW EARNS ITS PLACE. Left to habit, teams draw agent topologies that copy their own communication structure — a manager-agent because the team has a manager, a reviewer-agent because the team holds reviews, five negotiators because meetings have

five chairs. Conway observed the force; today insists it is a force to be *resisted*: the topology must copy the *information structure of the problem*, not the org chart of its authors. Badly coupled societies do not merely waste seats — measured error amplification in poorly partitioned multi-agent systems runs several-fold worse than in well-partitioned ones, because each agent’s mistake enters the others’ contexts and is repeated back with confidence, laundering error into consensus. An org chart is how *people* coordinate; an architecture is how *information* flows. When the two disagree, the org chart loses — or the system does.

The Dominance Criterion

THE SINGLE-VERSUS-MULTI-AGENT DEBATE NOW HAS A SCIENCE, and its conclusion is compact enough to be the day’s law. Across large controlled comparisons, multi-agent designs beat a single closed-loop agent *if and only if* the extra agents contribute **exogenous information** — knowledge the first agent could not reach: private constraints held by parties (the Potluck), or reading capacity beyond one faithful context (the Vacation). Where the first agent could already hold everything (the Symposium), the multi-agent variant not only fails to help; it amplifies error — badly coupled societies multiply one agent’s mistake through the transcript of every other, with measured amplification several times worse than well-partitioned designs. The “illusion of multi-agent advantage” in many published demos dissolves under this control: the win came from extra sampling or extra context, not from the society.

The criterion hands you a procedure, not just a verdict: before adding an agent, *name the information it brings that the current design lacks*. If you cannot name it, you are adding error surface with a org chart.

Pop Quiz — the Architect’s Reflex

A MULTIPLE-CHOICE QUESTION, WITH A SHOW OF HANDS. For the Symposium schedule — five speakers, eight printed constraints, one valid answer — which topology should a competent team reach for? **(a)** a star orchestrator with five speaker agents; **(b)** a five-agent debate over three rounds; **(c)** a hierarchical planner spawning worker agents. Choose one — then say back what information your chosen second agent brings that the first agent lacked.

Answer: none of them — and if the say-back clause left you empty-handed, that emptiness *was* the answer. Every constraint sat in the brief; no second

agent brings anything but error surface; one agent with a check loop wins, and twenty lines of exhaustive search wins harder. A team forced to pick from fashionable options is a model forced to answer from bad context: rejecting the options offered is itself the architecture. And yes — from this quiz onward, “none of the above” stays live in every question. Good. That standing vigilance is the point.

The Anatomy of Failure

MULTI-AGENT FAILURES ARE NOT RANDOM; they cluster, and the field’s failure taxonomies now name the clusters with the confidence of pathology. Three families cover most of the corpse-count, and today’s arenas were engineered so that each shows its face at least once:

Specification failures. The task was never routed to anyone; two agents each assumed the other owned a constraint; a role card’s authority was never granted. The Potluck’s unasked budget question lives here — the org chart had no arrow for the fact to travel.

Coordination failures. Consensus inertia; premature convergence; the polite committee that concedes a constraint to keep the peace; the tentative placement promoted to gospel because re-opening it felt rude. The Symposium’s negotiators supply the specimens.

Verification failures. The checker that blesses; the judge that trusts digests; the self-grading researcher drifting lenient; the rubric gamed into a loveless toast. Arenas Three and Four split this family between them.

Above them all sits the fatal move, which is a failure of *epistemics*: the moment a conclusion stops being questionable, the system’s remaining intelligence is spent decorating a wrong answer. The structural cures are the ones you practiced — verification that reaches back, ratification rounds, evidence demanded per claim, and records in which nothing unverified is ever load-bearing. And note the human mirror: your leak logs are this taxonomy collected live, one broken role at a time — when we read them aloud, file each leak into its family and the taxonomy will feel less like literature and more like Tuesday.

Act IV: The Architect's Procedure

EVERYTHING ABOVE COMPRESSES INTO A CATECHISM — the ordered questions that take you from a task statement to a defensible topology. Run them in order; refuse to skip.

1. **Where does the information live?** In the brief, in parties, in a rubric, in the world? This single question usually draws the topology by itself.
2. **What is checkable, and at what price?** Enumerate the verifiers available — mechanical, judged, human, the parties themselves — and their costs.
3. **What is the decomposition grain?** Can the task be cut until each step is trivially checkable? The finer the honest grain, the longer the horizon you may attempt.
4. **Where do the four organs sit?** Assembly, Proposal, Action, and Verification with its feedback — placed, fused, or delegated, but present and findable.
5. **Where is the fatal-move surface?** Which early conclusions become load-bearing? Who can un-believe them, and through what channel?
6. **What does the single-agent baseline score?** Build Rung 1 first. Every additional agent must beat it, and must name the exogenous information that justifies its seat.
7. **What is the error budget?** Steps, per-step reliability, amplification paths; where does the exponent bite, and which verifier blunts it?
8. **What is packaged as a skill?** Which verifiers and specialists should be contracts — named, tested, versioned — rather than vibes? (See below.)
9. **What are the stop rules?** Success, insolvency, futility, deference — the four exits, wired before the first run, with escalation paths that reach a human.

Skills as Load-Bearing Components

TWO OF THE MORNING'S WINNERS WERE SECRETLY SKILLS-WEEK LESSONS. The Toast's mechanical checker, married to its rubric, already *is* the executable heart of a skill: package it, and the same verification standard becomes portable — any generator, in any topology, on any team, held to the same eight checks. Verification stops being a box you redraw per project and becomes a component you ship, version, and test. And the Vacation's researcher points the other way: its failure mode was never self-grading per se but *unstructured* self-grading, and a well-written skill is the cure — a contract that takes the family's rubric, studies the page item by item, and returns *only* verdicts with quoted evidence. The skill is what makes a sub-agent trustworthy enough to delegate to, and what makes the judge's audit cheap, because evidence arrives pre-attached.

TO MAKE "SKILL AS CONTRACT" CONCRETE, here is the researcher skill's specification, small enough for an index card and strong enough to make fan-out safe:

Takes. One destination's page handle; the client's rubric, verbatim, as a checklist.

Does. Reads the page through the supplied tool only; walks the rubric item by item; never consults memory of other destinations; never fills a gap with a guess.

Returns. Per rubric item: *satisfied* / *violated* / *page silent*, each with a quoted sentence from the page — and nothing else. No prose, no enthusiasm, no ranking.

Refuses. Any question outside the rubric; any request to compare destinations — comparison is the judge's monopoly, and the contract says so.

Notice what the contract buys architecturally: the "page silent" verdict makes missing information *visible* instead of hallucinated; the quotation requirement makes the judge's audit a string search instead of an investigation; and the refusal clause enforces separation of duties at the component boundary rather than by exhortation. This is Postel's principle worn by a sub-agent: conservative in what it emits, suspicious of what it is asked.

The bridge sentence, worth saying twice: *topology decides who talks to whom; skills decide what a box is — a named, tested, versioned component*

instead of “some agent.” And a last hook for a future week: an imported skill is someone else’s verifier. Decide how much you trust it the way you would decide about any supply chain.

Instrumentation, Risks, and Plan B

INSTRUMENT THE BUILD ROUND LIKE THE PRODUCTION RUN IT IS RE-HEARSING. Every group logs every inter-agent message — the crib sheet shows the one-line hook — because the logs are the debrief’s evidence: the moment a wrong belief entered state, the round where a committee converged prematurely, the iteration where the toast’s structure should have died. A build that worked but cannot show *why* earns half credit; a build that failed and can point to the exact message where the fatal move occurred earns more than half, because the point of the morning is diagnosis, not victory. Cap everything that can run away: loop iterations, fan-out width, and — say it as engineering, not scolding — the token bill is part of the architecture.

THE KNOWN RISKS, WITH THEIR BRANCHES. The Wikipedia service misbehaving is already absorbed by the cache; if a page still errors, all groups share the cached snapshots and the comparison stays fair. A group with broken ADK credentials pairs into a neighbor’s build for Round 2 — they have already role-played; do not spend twenty minutes on keys. A role-play room that collapses into ordinary chatting gets its harness stiffened once (notes only, arrows only); if the topology is truly unworkable, allow one redraw — but it must be redrawn and re-signed, and the harness announces it as what it is: a redeploy, which costs the group two minutes of enactment time and is logged. A group that finishes the Symposium in ten minutes with code has *succeeded*; stretch them — make the program also explain the schedule from the constraints, then ask which part needed a model at all. And if the sprint overruns, cut the build round to forty-five minutes and the debrief to twenty-five by dropping one question per arena — but never the Potluck–Symposium whiplash, which is the spine of the day and survives no cut.

The Debrief Script

RUN THE DEBRIEF ARENA BY ARENA, three groups to a panel, sixty seconds of topology and result each, then the graded verdict, then one question per arena, asked to the room:

- *Potluck*: what did your plan violate — and at what moment was that violation already inevitable? (The moment the topology gave that fact no path to travel.)
- *Symposium*: what information did any second agent bring that the first lacked? (Silence, then the whiplash: P1's winner is P2's loser.)
- *Toast*: which checks did you route to code and which to a judge — and what did each iteration cost? (The cascade: free gates first, judgment last.)
- *Vacation*: who held the rubric — and what would a lazy researcher's verdict look like under each arrangement? (Separation of duties; the credulous judge.)

Close with the leak logs: read the three best entries aloud, match a Round-1 leak to a Round-2 silent failure wherever the rooms provide one, and let the room say the morning's map back — where the information lived, which topology won, where the verifier sat, and what the fatal move was — before the filled table goes up.

Arena	Information lives...	Winning topology	Verifier	Fatal move
Potluck	in the parties	star + elicitation	the guests ratify	menu before elicitation
Symposium	all in the brief	one agent + check	twenty lines of search	tentative → gospel
Toast	in the rubric	generator-verifier	script first, judge last	structure locked early
Vacation	out in the world	fan-out + judge	veto, quotes only	digest as ground truth

The Take-Home

TWO SMALL ASSIGNMENTS CARRY THE DAY INTO THE WEEK. First, the *redesign*: take your own Round-1 topology drawing — the one the harness signed and enforced — and, with the verdicts in hand, redraw it. One page: the new drawing, the four organs labeled, and a paragraph naming what you moved and which verdict moved it. Bring both drawings; the diff is the deliverable. Second, the *leak-log postmortem*: for each entry in your group's leak log, file it into the failure taxonomy — specification, coordination, or verification — and state the structural

change (an arrow, a box, a contract) that would have made the leak unnecessary. A leak with no structural cure is worth a margin note of its own: it usually means the human was compensating for a missing verifier, and those are the most expensive absences to discover in production instead of in a classroom.

Glossary of the Day

Architecture. The arrangement of loops: how many, which organs they share, whose verification may overwrite whose state.

Open loop. Propose, act, never check; success p^n ; a gesture at scale.

Fatal move. A wrong early conclusion made load-bearing with no return path; $P_{\text{success}} = 0$ from that moment.

Four organs. Assembly, Proposal, Action, Verification-with-feedback — present in every honest loop, findable on every honest drawing.

Elicitation. The design of questions as architecture; the only route to information held by parties who deflect.

Ratification round. Re-presenting a plan to the parties who hold the constraints; verification by the sources themselves.

Code-as-orchestration. Control flow written in a programming language, the model invited in only where judgment is needed; the Symposium's honest champion.

Generator-verifier loop. Generate, gate, feed failures back by name, repeat; the arithmetic that turns 35% into 80%-plus.

Fan-out / reduce. Parallel clean-room workers over a partition; one skeptical reducer with the global view.

Context partitioning. The unromantic truth of much "multi-agent": a filing discipline, not a society.

Exogenous information. Knowledge the current design cannot reach; the only honest purchase price of an additional agent.

Error amplification. One agent's mistake entering others' contexts and returning as consensus; several-fold worse in badly partitioned societies.

Self-conditioning. A transcript's errors making further errors likelier; the open loop's hazard growing as it tires.

Leak log. The role-play harness's record of broken roles; a human-run audit of where a topology starves its agents.

Skill (as component). A named, tested, versioned contract — what a box *is*, while topology says who talks to whom.

The Afternoon Runsheet

THE AFTERNOON IS A LECTURE WITH THE MORNING AS ITS EXAMPLE SET. Its deck opens with the same provocation the day opened with — now with the room's own experience standing behind every slide — and unseals the verdicts arena by arena. The intended rhythm:

Segment	Approx.	Content
Re-entry & verdicts	45 min	the four arenas judged; three groups compared per arena
The catalog	40 min	ladder, workhorses, adaptive fringe, partitioning reframe
Break	15 min	
The science	45 min	calculus, horizons, dominance criterion, anatomy of failure
The lineage	20 min	von Neumann 1956; TCP over IP; the provocation answered
The procedure	25 min	the nine questions; skills as components; say-it-back recap
Close	10 min	the traveled map; the law; pointers into Archcraft

THREE STAGING NOTES. Keep the Symposium verdict *immediately after* the Potluck verdict — the whiplash between them is the deepest single beat of the day and survives no padding. Hold the von Neumann and TCP slides until after the science, where they land as lineage rather than trivia. And end the say-it-back recap with the room, not the lecturer, reconstructing the four-arena map — the filled table goes up only after the room has said it back.

Where to Go Deeper

TODAY'S MATERIAL COMPRESSES THE ARCHCRAFT MONOGRAPH, which remains the reference of record for this week: the reliability calculus and the verifier arithmetic in its early chapters; the workhorse and adaptive pattern catalogs; the science of scaling and the dominance criterion; the anatomy-of-failure taxonomy; the harness turn and the protocol layer; and the comparative anatomy of three production harnesses opened on the bench. Students who want one sitting's worth: read the reliability chapter and the synthesis chapter, in that order, and

bring your objections on paper — the ones the morning's arenas cannot already answer are the interesting ones.

A Calibration Note

THIS SESSION RUNS ON A DELIBERATE RHYTHM: feel first, name after. The morning's arenas are engineered so that the afternoon's catalog arrives as recognition, not memorization — the star, the pipeline, the loop, and the fan-out will each wear the face of something your own group built, botched, or watched a neighbor botch at eleven in the morning. Protect the debrief time; it is where the experiences become transferable. And keep the provocation's resolution for the very end, where it lands with the whole day behind it: the arithmetic of p^n was never wrong — it simply describes a machine nobody should build. Decompose until steps are checkable; verify so that nothing unverified is ever load-bearing; let the loop reach back and un-believe. That is how a million steps survive, how a packet crosses an unreliable planet intact, and how sixty fallible components — some of them silicon, some of them us — taught one another architecture today.

Where does the information live? Ask that, close the loop, guard against the fatal move — and the topology draws itself. Extra agents are purchased with exogenous information, and with nothing else.

Next week we take the architectures we can now name and ask what it costs to run them honestly — and what it takes to make one improve itself. Bring your leak logs; they are not done teaching.

There were 27 pages in this document.

FEEDBACK

Please send any feedback on this document, or report errors to the author at asif@supportvectors.com